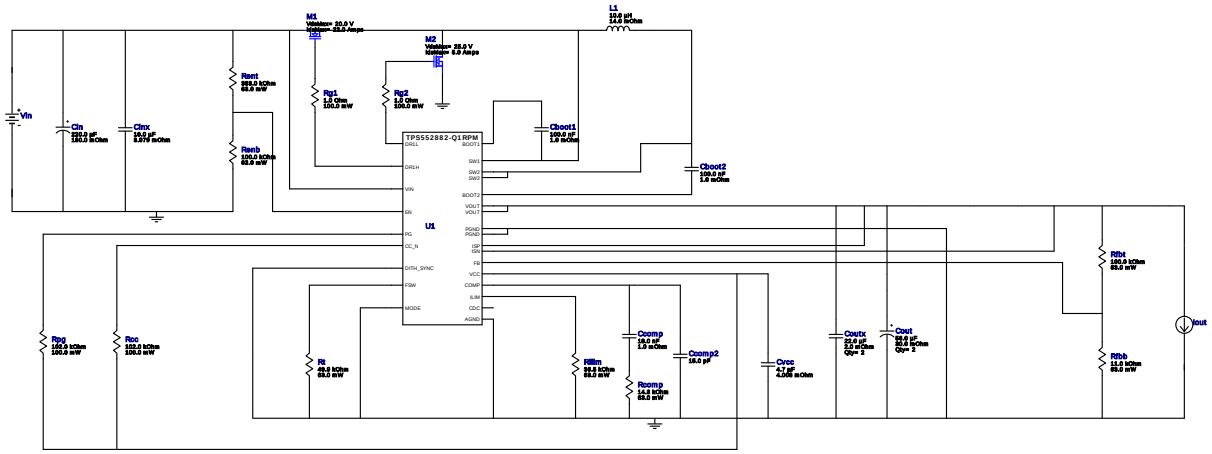
WEBENCH® Design Report

VinMin = 6.0V
VinMax = 18.0V
Vout = 12.0V
Iout = 3.0A

Device = TPS552882QRPMRQ1
Topology = Buck_Boost
Created = 2026-10-11 15:22:30.185
BOM Cost = $6.84
BOM Count = 28
Total Pd = 2.87W

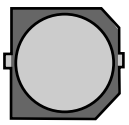
Design : 18 TPS552882QRPMRQ1
TPS552882QRPMRQ1 6V-18V to 12.00V @ 3A



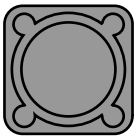
Design Alerts

Component Selection Information
The TPS552882-Q1 is qualified for Automotive applications. All passives and other components selected in this design may not be qualified for Automotive applications. The user is required to verify that all components in the design meet the qualification and safety requirements for their specific application.

Electrical BOM

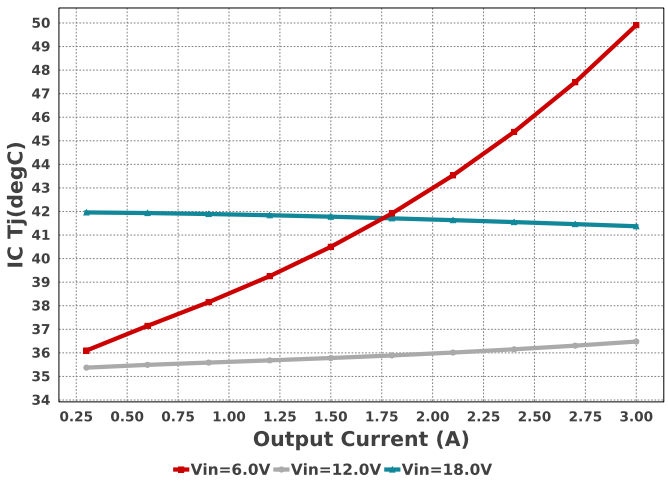
Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cboot1	Taiyo Yuden	EMK107B7104KA-T Series= X7R	Cap= 100.0 nF ESR= 1.0 mOhm VDC= 16.0 V IRMS= 0.0 A	1	$0.01	0603 5 mm^2
Cboot2	Taiyo Yuden	EMK107B7104KA-T Series= X7R	Cap= 100.0 nF ESR= 1.0 mOhm VDC= 16.0 V IRMS= 0.0 A	1	$0.01	0603 5 mm^2
Ccomp	MuRata	GRM155R71C183KA01D Series= X7R	Cap= 18.0 nF ESR= 1.0 mOhm VDC= 16.0 V IRMS= 0.0 A	1	$0.01	0402 3 mm^2
Ccomp2	Kemet	C0402C150J4GACTU Series= C0G/NP0	Cap= 15.0 pF VDC= 16.0 V IRMS= 0.0 A	1	$0.01	0402 3 mm^2
Cin	Nichicon	UUD1H221MNL1GS Series= uD	Cap= 220.0 uF ESR= 180.0 mOhm VDC= 50.0 V IRMS= 670.0 mA	1	$0.30	 SM_RADIAL_10BMM 160 mm^2
Cinx	Taiyo Yuden	MSASU32MSB5106KPNA01 Series= X5R	Cap= 10.0 uF ESR= 3.979 mOhm VDC= 50.0 V IRMS= 3.4821 A	1	$0.16	1210 15 mm^2
Cinx2	TDK	C1005X5R1E105K050BC Series= X5R	Cap= 1.0 uF ESR= 11.4 mOhm VDC= 25.0 V IRMS= 0.0 A	1	$0.02	0402 3 mm^2

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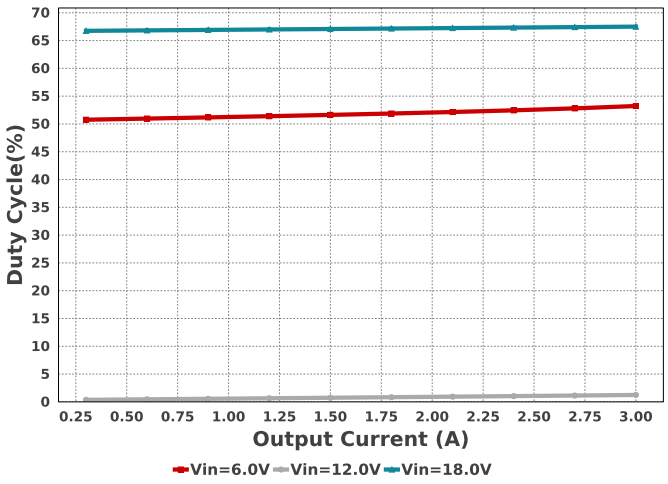
Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cout	Panasonic	20SVPF56MX Series= SVPF	Cap= 56.0 uF ESR= 30.0 mOhm VDC= 20.0 V IRMS= 2.8 A	2	$0.47	 CAPSMT_62_E61 53 mm²
Coutx	MuRata	GRM32ER61E226KE15L Series= X5R	Cap= 22.0 uF ESR= 2.0 mOhm VDC= 25.0 V IRMS= 3.67 A	2	$0.23	 1210 15 mm²
Coutx2	TDK	C1005X5R1E105K050BC Series= X5R	Cap= 1.0 uF ESR= 11.4 mOhm VDC= 25.0 V IRMS= 0.0 A	1	$0.02	 0402 3 mm²
Cvcc	TDK	C1608X5R1C475K080AC Series= X5R	Cap= 4.7 uF ESR= 4.008 mOhm VDC= 16.0 V IRMS= 0.0 A	1	$0.06	 0603 5 mm²
L1	Coilcraft	MSS1210-103MEB	L= 10.0 µH 14.0 mOhm	1	$0.81	 MSS1210 204 mm²
M1	Texas Instruments	CSD15571Q2	VdsMax= 20.0 V IdsMax= 22.0 Amps	1	$0.08	DQK0006C 9 mm²
M2	Texas Instruments	CSD16301Q2	VdsMax= 25.0 V IdsMax= 5.0 Amps	1	$0.11	DQK0006C 9 mm²
Rcc	Vishay-Dale	CRCW0603102KFKEA Series= CRCW..e3	Res= 102.0 kOhm Power= 100.0 mW Tolerance= 1.0%	1	$0.01	 0603 5 mm²
Rcomp	Vishay-Dale	CRCW040214K3FKED Series= CRCW..e3	Res= 14.3 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	 0402 3 mm²
Renb	Vishay-Dale	CRCW0402100KFKED Series= CRCW..e3	Res= 100.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	 0402 3 mm²
Rent	Vishay-Dale	CRCW0402383KFKED Series= CRCW..e3	Res= 383.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	 0402 3 mm²
Rfbb	Vishay-Dale	CRCW040211K0FKED Series= CRCW..e3	Res= 11.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	 0402 3 mm²
Rfbt	Vishay-Dale	CRCW0402100KFKED Series= CRCW..e3	Res= 100.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	 0402 3 mm²
Rg1	Vishay-Dale	CRCW06031R00FKEA Series= CRCW..e3	Res= 1.0 Ohm Power= 100.0 mW Tolerance= 1.0%	1	$0.01	 0603 5 mm²
Rg2	Vishay-Dale	CRCW06031R00FKEA Series= CRCW..e3	Res= 1.0 Ohm Power= 100.0 mW Tolerance= 1.0%	1	$0.01	 0603 5 mm²
Rilim	Yageo	AC0402FR-0736K5L Series= ?	Res= 36.5 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	 0402 3 mm²
Rpg	Vishay-Dale	CRCW0603102KFKEA Series= CRCW..e3	Res= 102.0 kOhm Power= 100.0 mW Tolerance= 1.0%	1	$0.01	 0603 5 mm²
Rt	Vishay-Dale	CRCW040249K9FKED Series= CRCW..e3	Res= 49.9 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	 0402 3 mm²
U1	Texas Instruments	TPS552882QRPMRQ1	Switcher	1	$3.76	RPM0026A-MFG 22 mm²

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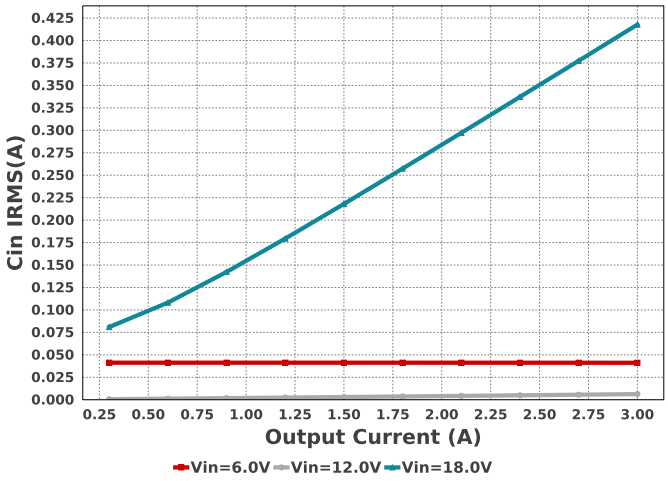
IC Tj



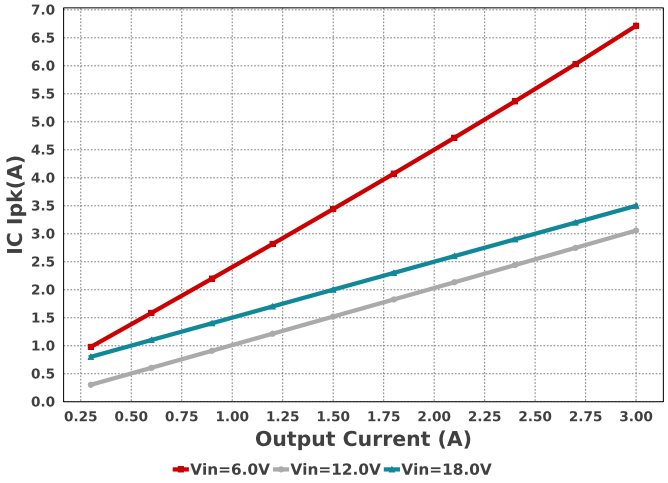
Duty Cycle



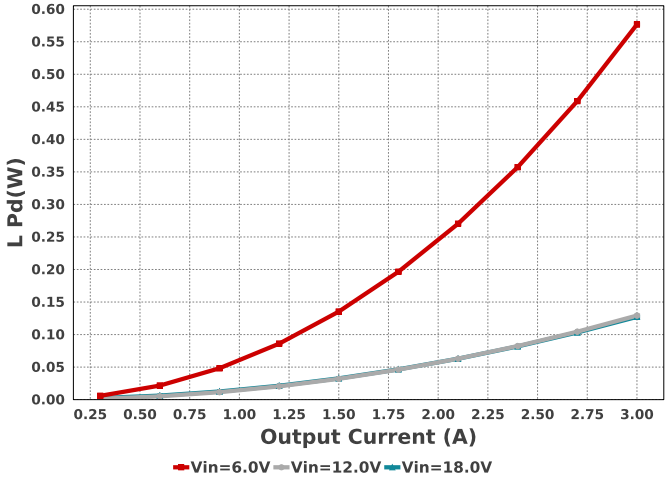
Cin IRMS



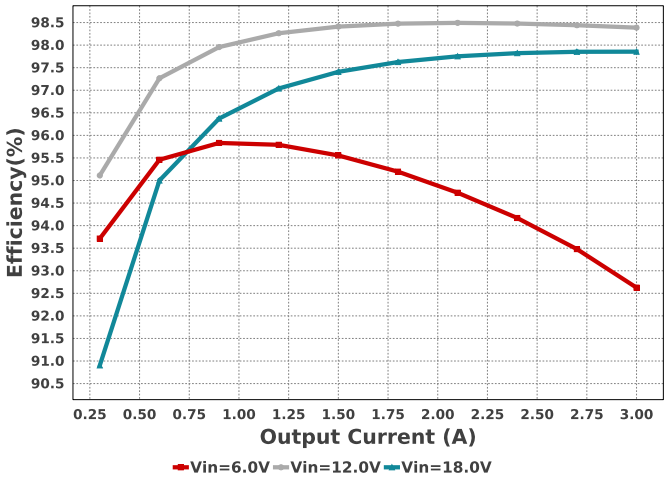
IC Ipk



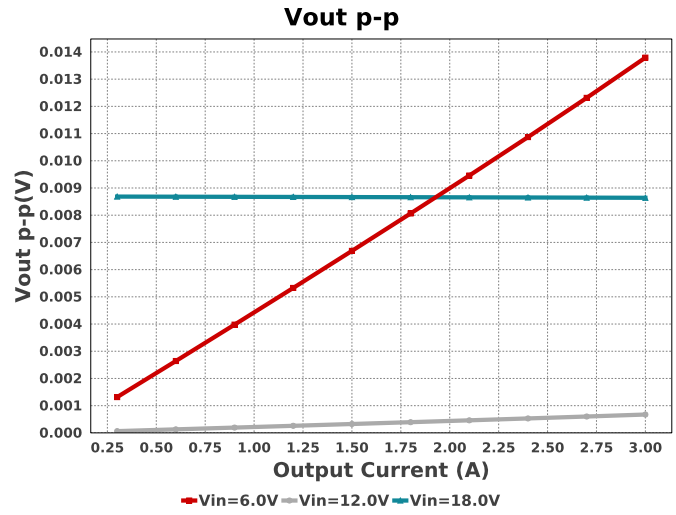
L Pd

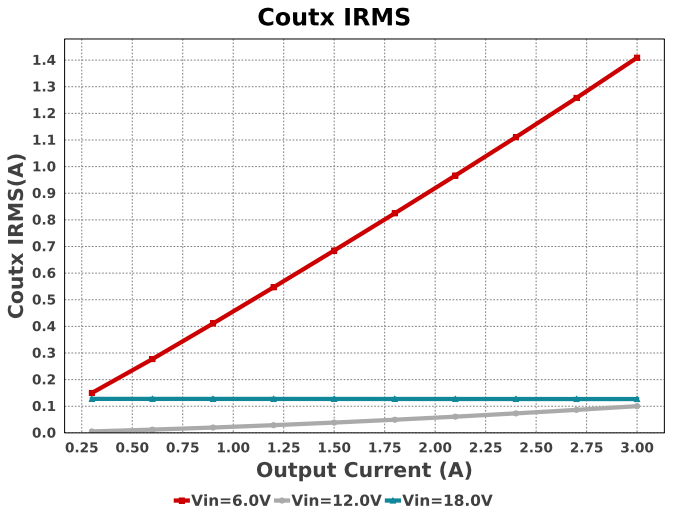


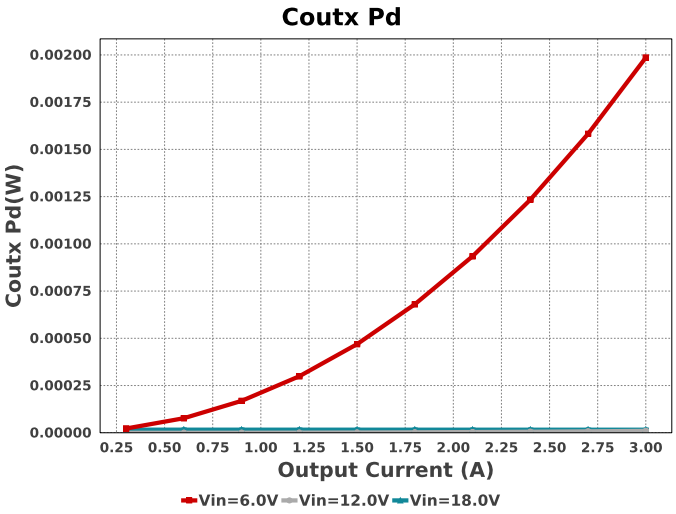
Efficiency

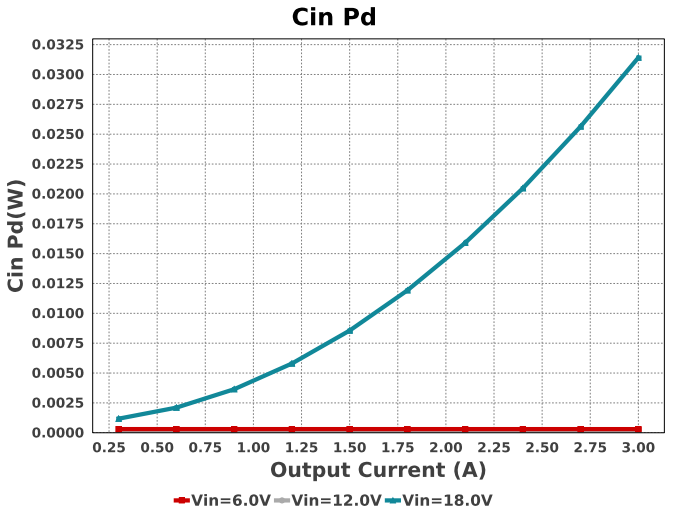


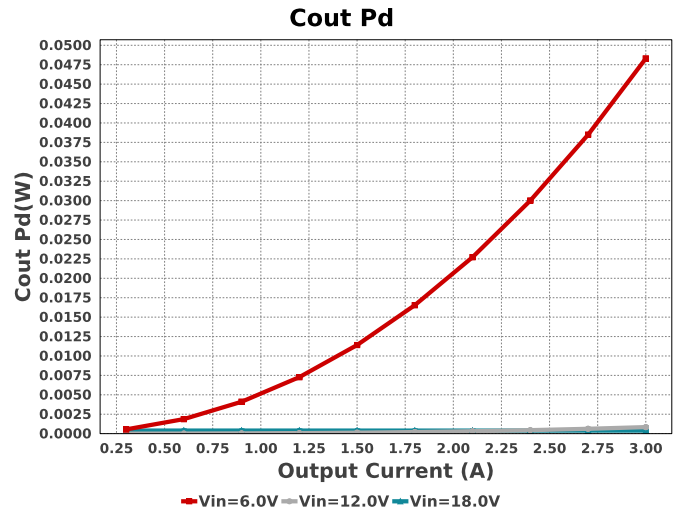
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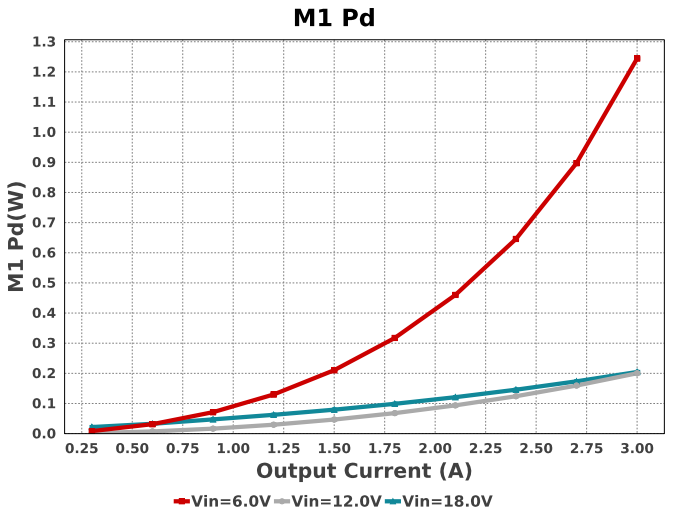




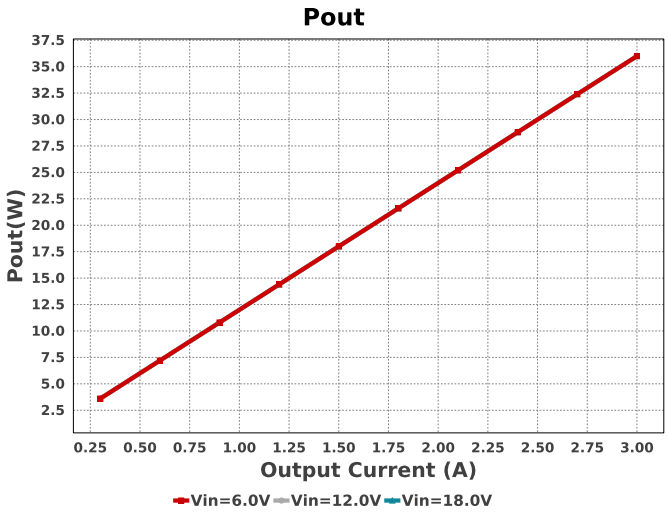


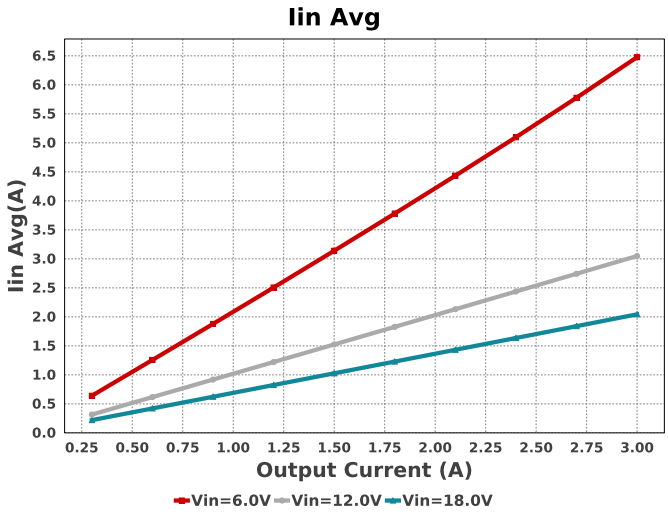


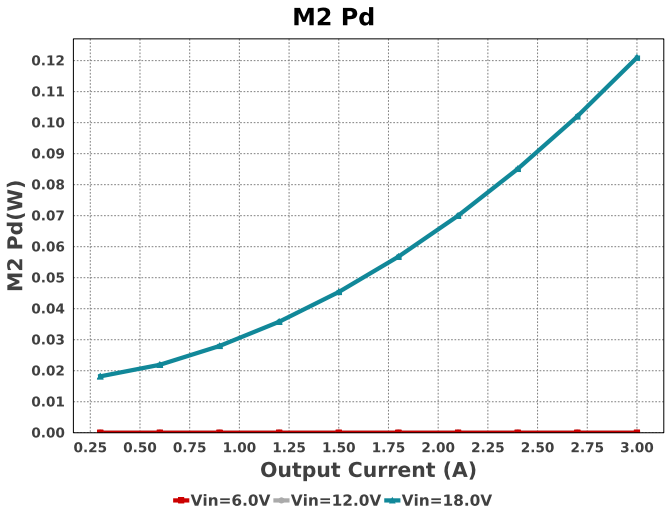


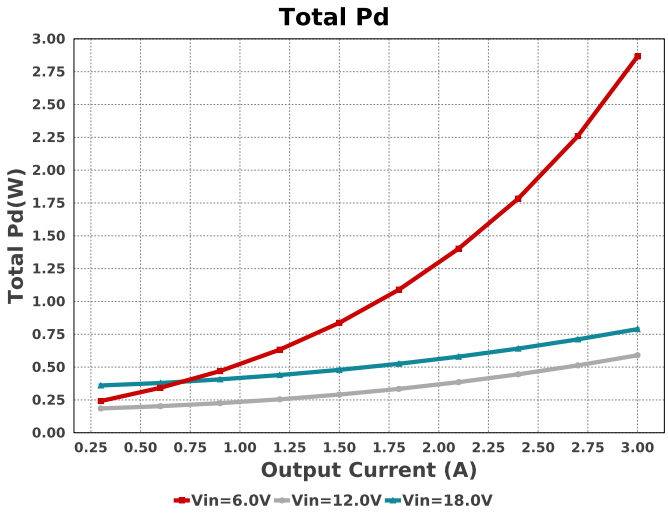


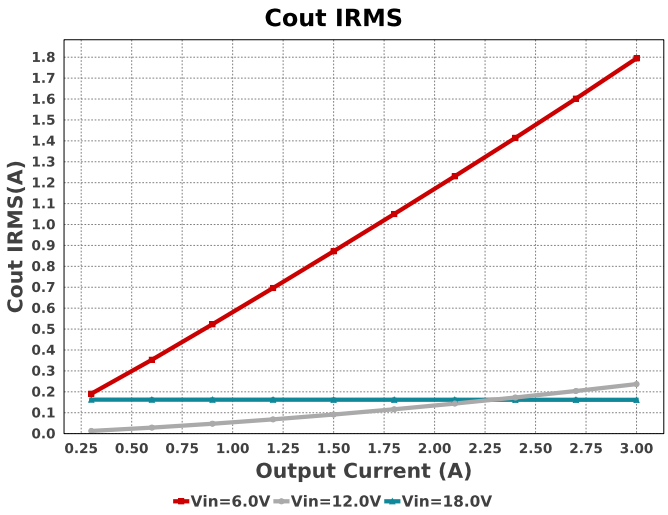
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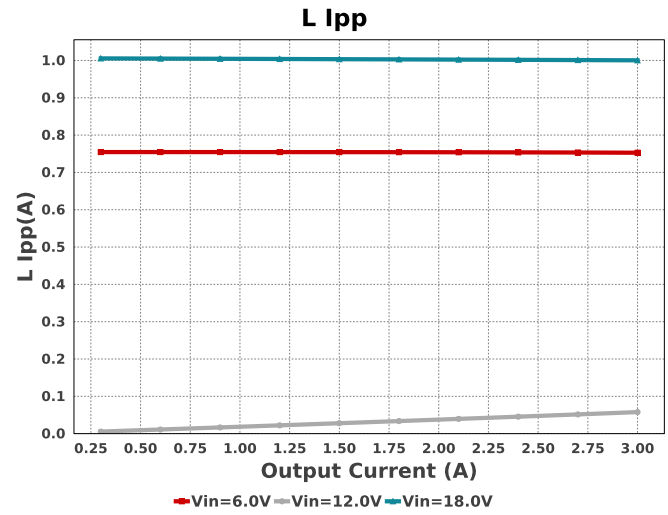




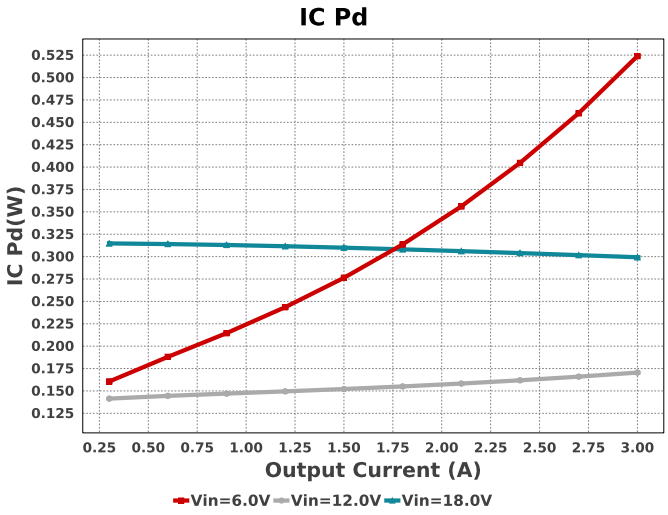


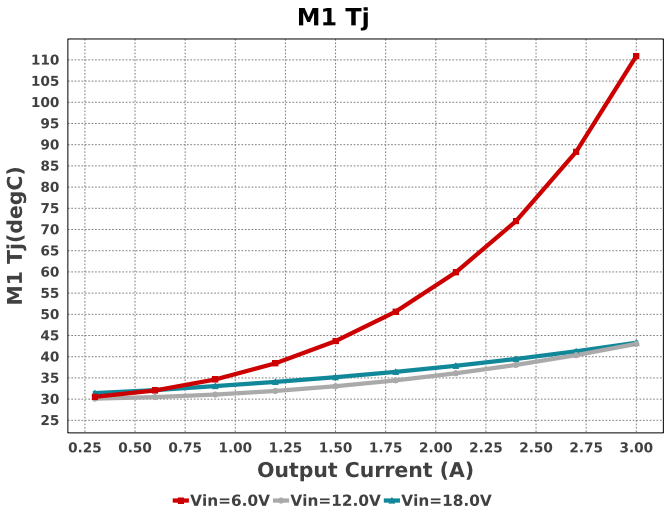


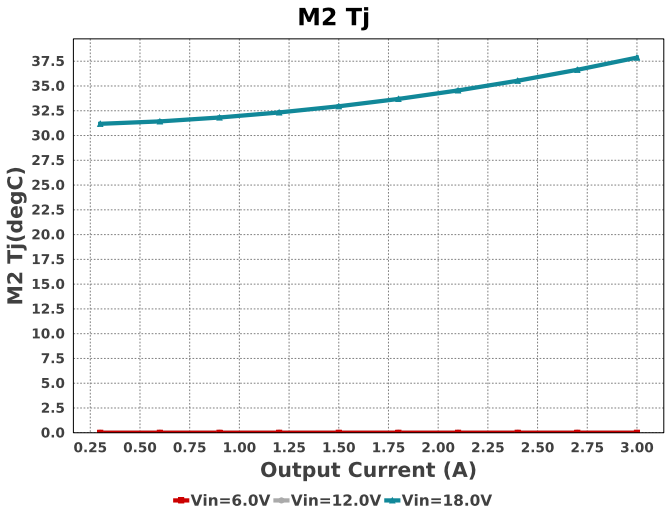


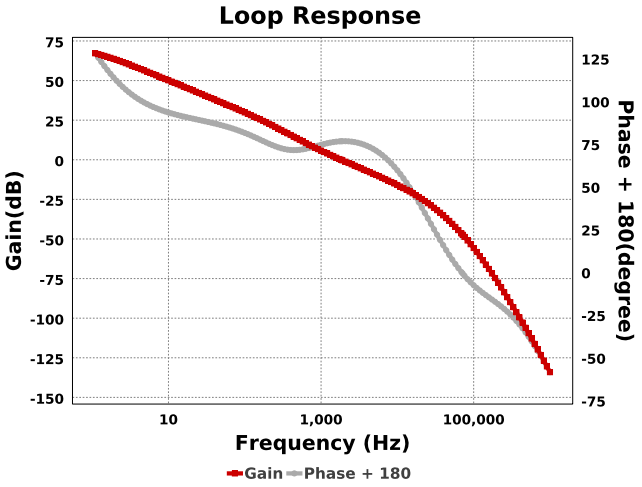


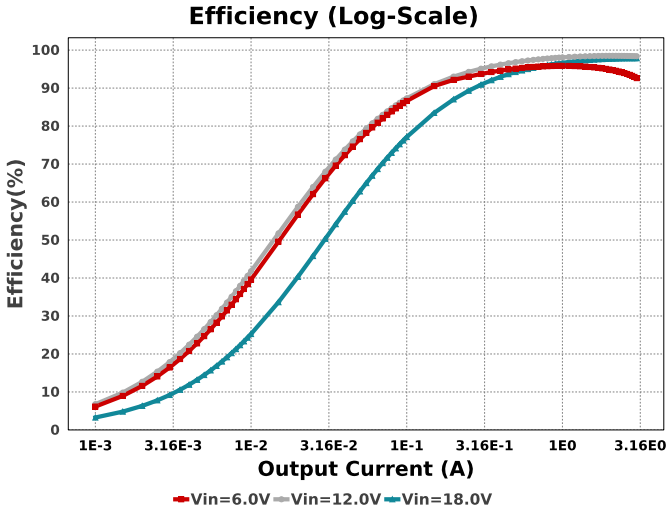
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Operating Values

#	Name	Value	Category	Description
1.	Cin IRMS	41.182 mA	Capacitor	Input capacitor RMS ripple current
2.	Cin Pd	305.27 µW	Capacitor	Input capacitor power dissipation
3.	Cout IRMS	1.794 A	Capacitor	Output capacitor RMS ripple current
4.	Cout Pd	48.299 mW	Capacitor	Output capacitor power dissipation
5.	Coutx IRMS	1.409 A	Capacitor	Output capacitor_x RMS ripple current
6.	Coutx Pd	1.986 mW	Capacitor	Output capacitor_x power loss
7.	IC Ipk	6.715 A	IC	Peak switch current in IC
8.	IC Pd	523.94 mW	IC	IC power dissipation
9.	IC Tj	49.91 degC	IC	IC junction temperature
10.	IC Tolerance	12.0 mV	IC	IC Feedback Tolerance
11.	ICThetaJA	38.0 degC/W	IC	IC junction-to-ambient thermal resistance

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#	Name	Value	Category	Description
12.	Iin Avg	6.478 A	IC	Average input current
13.	L Ipp	752.73 mA	Inductor	Peak-to-peak inductor ripple current
14.	L Pd	576.55 mW	Inductor	Inductor power dissipation
15.	M1 Pd	1.244 W	Mosfet	M1 MOSFET total power dissipation
16.	M1 Tj	110.89 degC	Mosfet	M1 MOSFET junction temperature
17.	M2 Pd	0.0 W	Mosfet	M2 MOSFET total power dissipation
18.	M2 Tj	0.0 degC	Mosfet	M2 MOSFET junction temperature
19.	Cin Pd	305.27 µW	Power	Input capacitor power dissipation
20.	Cout Pd	48.299 mW	Power	Output capacitor power dissipation
21.	Coutx Pd	1.986 mW	Power	Output capacitor_x power loss
22.	IC Pd	523.94 mW	Power	IC power dissipation
23.	L Pd	576.55 mW	Power	Inductor power dissipation
24.	M1 Pd	1.244 W	Power	M1 MOSFET total power dissipation
25.	M2 Pd	0.0 W	Power	M2 MOSFET total power dissipation
26.	Total Pd	2.866 W	Power	Total Power Dissipation
27.	BOM Count	28	System Information	Total Design BOM count
28.	Cross Freq	867.533 Hz	System Information	Bode plot crossover frequency
29.	Duty Cycle	53.225 %	System Information	Duty cycle
30.	Efficiency	92.626 %	System Information	Steady state efficiency
31.	FootPrint	621.0 mm^2	System Information	Total Foot Print Area of BOM components
32.	Frequency	397.614 kHz	System Information	Switching frequency
33.	Gain Marg	-24.411 dB	System Information	Bode Plot Gain Margin
34.	Iout	3.0 A	System Information	Iout operating point
35.	Low Freq Gain	55.155 dB	System Information	Gain at 1Hz
36.	Mode	CCM	System Information	Conduction Mode
37.	Phase Marg	87.025 deg	System Information	Bode Plot Phase Margin
38.	Pout	36.0 W	System Information	Total output power
39.	Total BOM	$6.84	System Information	Total BOM Cost
40.	Vin	6.0 V	System Information	Vin operating point
41.	Vout	12.0 V	System Information	Operational Output Voltage
42.	Vout Actual	12.109 V	System Information	Vout Actual calculated based on selected voltage divider resistors
43.	Vout Tolerance	2.838 %	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
44.	Vout p-p	13.79 mV	System Information	Peak-to-peak output ripple voltage

Design Inputs

Name	Value	Description
Iout	3.0	Maximum Output Current
VinMax	18.0	Maximum input voltage
VinMin	6.0	Minimum input voltage
Vout	12.0	Output Voltage
base_pn	TPS552882-Q1	Base Product Number
source	DC	Input Source Type
Ta	30.0	Ambient temperature

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WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of Cin and Cout, and the inductance and DC resistance of L1 before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

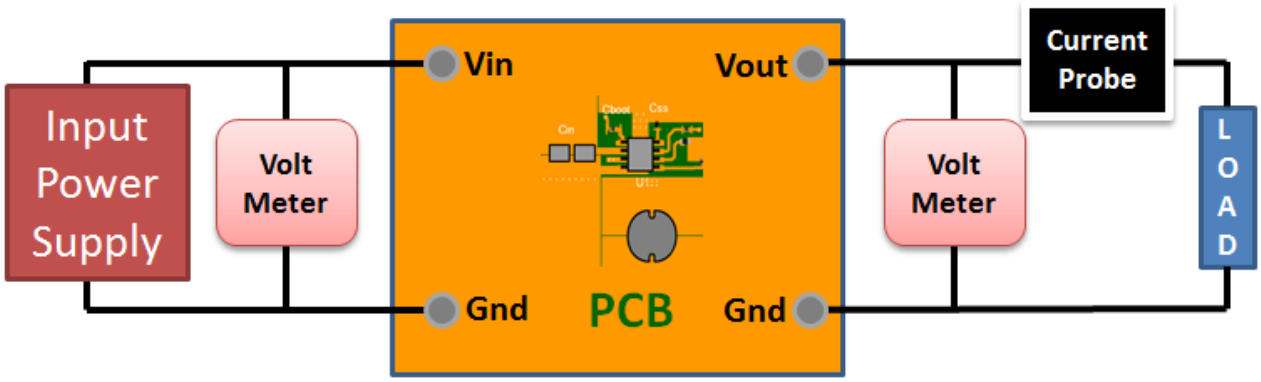
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab town to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 6.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to Vin and GND. Connect a digital volt meter and a load if needed to set the minimum Iout of the design from Vout and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between Vin and GND, a load is connected between Vout and GND and a current meter is connected in series between Vout and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



Design Assistance

1. Master key : D7800E4DB5C3DF729D7B162E4F148F49[v1]
2. **TPS552882-Q1** Product Folder : http://www.ti.com/product/TPS552882%2DQ1 : contains the data sheet and other resources.

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